

Bibin Alias

Embedded Systems & Firmware Engineer

- Phone: +91 85905 50607
- Email: bibinalias1@gmail.com
- Website: <https://bibinalias.github.io>
- LinkedIn: <https://www.linkedin.com/in/bibinalias>
- Address: Bangalore, India – 560095

Summary:

Engineering professional with 6+ years of hands-on experience designing and developing complex embedded software, firmware, and low-level device drivers using Embedded C. Proven expertise across wireless networking protocols (Wi-SUN, NB-IoT, Bluetooth, Wi-Fi, Z-Wave), RTOS, and Embedded Linux. Experienced in Python-based automated testing, MVC architecture, and hardware board bring-up. Strong track record of executing complex projects under tight deadlines while ensuring compliance and high reliability.

Technical Skills:

- **Languages & Scripting:** Embedded C, Python, Bash, MicroPython, Assembly, MATLAB, HDL
- **Operating Systems & RTOS:** FreeRTOS, Embedded Linux (Yocto), Bare-Metal
- **Protocols & Wireless:** Wi-SUN, NB-IoT, Bluetooth (GATT, GAP, HCI), Wi-Fi, Z-Wave, Zigbee, MQTT/MQTTS, OpenTherm, CAN, I2C, SPI, QSPI, UART, RS-485, RS-232, TCP/UDP, D-Bus
- **Microcontrollers & Hardware:** STM32 (STM32H563, STM32F091RC), ARM Cortex-M0/M4/M33, i.MX 8M Mini, ZG28, MGM210P, ZGM130S, Raspberry Pi, BeagleBone Black, Arduino
- **Firmware & Development:** Firmware Development, Device Drivers, Board Bring-Up, Debugging (JTAG/SWD), Unit Testing
- **AI-Assisted Engineering:** GitHub Copilot and LLM-based tools for accelerated code design & development, automated unit test generation, and technical documentation
- **IDE & Tools:** Eclipse, Simplicity Studio, STM32CubeMX, Keil, Coverity, Wireshark, Git, Gerrit, Jira, Qt Designer, OrCAD Capture
- **Hardware Testing & Lab:** Digital Storage Oscilloscope, Logic Analyzer, Arbitrary Waveform Generator, OpenTherm Certification Tool, 5S Standardized Laboratory Practices

Professional Experience:

Senior Software Engineer — Bosch Global Software Technologies, Bangalore

November 2023 – Present

- Designed, implemented, and tested the OpenTherm Plus (Slave) protocol for Bosch HVAC systems.
- Contributed enhancements to the OpenTherm Master protocol, since the MH220 also serves as a gateway and controlling device.
- Led OpenTherm certification testing for both Master and Slave protocols, achieving a 100% pass rate and ensuring full compliance with industry standards.
- Conducted unit testing and prototype validation to guarantee robust and reliable protocol execution.
- Developed Smart Meter Network Interface Card (NIC) applications utilizing TCP client/server architecture over Wi-SUN wireless mesh technology.
- Implemented the Wi-SUN application layer and constructed a custom UDP application for diagnostic and monitoring purposes.
- Configured Raspberry Pi border routers to establish IPv6 connectivity across Wi-SUN end nodes using Linux D-Bus inter-process communication.
- Designed and deployed secure MQTTS applications for NB-IoT modules.
- Performed Decision Analysis and Resolution (DAR) evaluations and project estimation across critical engineering deliverables.

Projects at Bosch:

- **HVAC MH220 Modules (STM32H563, embOS):** Firmware development enabling seamless communication between appliances/heat pumps and HMI units for building climate control.
- **Smart Meter NIC (ZG28, FreeRTOS):** Developed NIC firmware providing internet connectivity to smart meters via Border Routers for periodic Head End System (HES) and Network Monitoring System (NMS) telemetry.

Software Engineer — Thinkpalm Technologies Pvt. Ltd., Kochi

November 2022 – October 2023

- Developed, maintained, and debugged Bluetooth and WLAN device drivers and firmware across various host platforms and wireless chipsets.
- Analyzed Bluetooth/WLAN subsystem source code flow and managed protocol stacks including GATT, GAP, and HCI.
- Leveraged Coverity static analysis and Wireshark protocol analyzer to detect, investigate, and resolve firmware/driver bugs, significantly enhancing system stability.
- Modified and configured hostapd user-space daemon to ensure seamless integration and optimal performance with Wi-Fi chipsets.
- Participated in structured code reviews using Git and Gerrit across multi-vendor codebases.
- Received team recognition award for outstanding contribution and commitment to delivery excellence.

Projects at Thinkpalm:

- **Bluetooth & Wi-Fi Chipset Drivers** (*Linux, Bluetooth/Wi-Fi Chipsets*): Driver/firmware maintenance, bug resolution, and feature integration for host platforms.

Software Engineer — Gadgeon Smart Systems Pvt. Ltd., Kochi

April 2021 – November 2022

- Designed and developed Linux applications for MQTT (libmosquitto, mDNS), Zigbee, and Bluetooth (bluez) interfaces.
- Built a Network Co-Processor (NCP) application to enable Z-Wave communication for IoT gateways.
- Developed low-level Linux SPI device drivers for analog-to-digital converter (ADC) chips.
- Executed full board bring-up and hardware validation for HVAC controllers using logic analyzers and digital oscilloscopes.
- Designed and built a Model-View-Controller (MVC) automated Test Fixture Application using Python and PyQt5 for PCB component verification.
- Built custom Linux images and packages using Embedded Linux (Yocto Project).
- Collaborated with cross-functional teams to author technical documentation, software specs, and troubleshooting guides.

Projects at Gadgeon:

- **HVAC System with IoT** (*i.MX 8M Mini, MGM210P, ZGM130S, Embedded Linux*): Fan-coil firmware development and IoT stack integration.
- **Automated PCB Test Jig** (*Open-Q 845 uSOM, Linux, ARM Cortex-M0/M4, DAQ*): Test fixture for drone PCB board validation and functional diagnostics.

Firmware Engineer — Transight Systems, Kochi

November 2019 – April 2021

- Architected firmware for ARM Cortex-M0/M4 microcontrollers using FreeRTOS 10, bare-metal C, and Open CPU architecture.
- Designed and implemented low-level device drivers for Flash Memory (QSPI) and Inertial Measurement Units (IMU via I2C).
- Integrated and validated industrial communication protocols including CAN, I2C, QSPI, RS-485, and RS-232.
- Developed custom algorithms using IMU sensor data to detect driving events (harsh acceleration, hard braking, sharp turns, roll-over) for AIS 140 compliance.
- Built Python automated test tools for hardware/firmware validation, achieving Automotive Research Association of India (ARAI) certification.
- Integrated 2G/4G cellular telemetry modules and GPS receivers for real-time fleet positioning and diagnostics.
- Performed reverse engineering, schematic reviews (OrCAD), and board bring-up using oscilloscopes.

Projects at Transight:

- **Fleet Management Solution** (*ARM Cortex-M0/M4, RTOS, Bare-Metal, Open CPU*): Real-time GPS vehicle tracking system monitoring driver behavior, fuel metrics, and route telemetry.

Education:

- **Bachelor of Technology in Electronics and Communication Engineering**
Mar Athanasius College of Engineering – Mahatma Gandhi University (2015 – 2018)
- **Diploma in Electronics Engineering**
Board of Technical Education, Kerala (2012 – 2015)

Certifications & Workshops:

- IoT: Converging Technologies for Smart Environment National Workshop (2017)
- IoT with Microsoft Azure (2017)
- Entrepreneurship Awareness Camp (2016)
- Metabotics Workshop (2015)
- Robotic Workshop — National Robotic Championship (2014)
- High Power Transmitter Station Training — Doordarshan (2014)